In [2]:
```
#CIA3 - Continuation Of CIA2 (Paper Moisture Test)
#A.Deepak (2460601)
#16-02-2026

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load Data
df = pd.read_csv('paper_moisture_control.csv')
# 2. Timestamp Alignment
time_cols = ['timestamp', 'cycle_start', 'cycle_end']
for col in time_cols:
    df[col] = pd.to_datetime(df[col])
# 3. Data Cleansing (Create df_clean)
# Remove duplicates
df_clean = df.drop_duplicates().copy()
# Fill missing moisture values (Interpolation)
df_clean['moisture_pct'] = df_clean['moisture_pct'].ffill().interpolate(method='linear')
# Handle missing values in other columns (Drop rows where essential data is missing for simplicity)
df_clean.dropna(subset=['paper_grade', 'machine_id', 'temperature_C', 'humidity_pct'], inplace=True)
print(df_clean.head())
```

```
            timestamp machine_id   batch_id paper_grade  moisture_pct  \
0 2025-09-01 07:00:00       M004  BATCH_001      coated      6.229510
1 2025-09-01 07:10:00       M004  BATCH_001      coated      6.525639
2 2025-09-01 07:20:00       M004  BATCH_001      coated      6.528640
3 2025-09-01 07:30:00       M004  BATCH_001      coated      6.439983
4 2025-09-01 07:40:00       M004  BATCH_001      coated      7.116764

   temperature_C  humidity_pct         cycle_start           cycle_end  \
0      28.048834     56.041926 2025-09-01 07:00:00 2025-09-01 09:00:00
1      24.146738     54.039701 2025-09-01 07:00:00 2025-09-01 09:00:00
2      22.919657     54.670740 2025-09-01 07:00:00 2025-09-01 09:00:00
3      29.414033     33.000346 2025-09-01 07:00:00 2025-09-01 09:00:00
4      23.824238     45.115315 2025-09-01 07:00:00 2025-09-01 09:00:00

   deviation_flag
0               0
1               0
2               0
3               0
4               0
```

In [3]:
```python
from sklearn.preprocessing import LabelEncoder, MinMaxScaler, StandardScaler

#We have completed the first 3 steps. now the rest few steps are to be completed.

#4.Feature Engineering and Encoding

# Encode 'paper_grade' and 'machine_id'
le = LabelEncoder()
df_clean['paper_grade_enc'] = le.fit_transform(df_clean['paper_grade'])
df_clean['machine_id_enc'] = le.fit_transform(df_clean['machine_id'])

# Check the new columns
print(df_clean[['paper_grade', 'paper_grade_enc', 'machine_id', 'machine_id_enc']].head())

# Normalization
min_max_scaler = MinMaxScaler()
df_clean['moisture_minmax'] = min_max_scaler.fit_transform(df_clean[['moisture_pct']])
std_scaler = StandardScaler()
df_clean['moisture_zscore'] = std_scaler.fit_transform(df_clean[['moisture_pct']])
```

```
  paper_grade  paper_grade_enc machine_id  machine_id_enc
0      coated                0       M004               3
1      coated                0       M004               3
2      coated                0       M004               3
3      coated                0       M004               3
4      coated                0       M004               3
```

In [4]:
```
from sklearn.preprocessing import MinMaxScaler, StandardScaler

#4.1. Min-Max Scaling (Values between 0 and 1)
min_max_scaler = MinMaxScaler()
df_clean['moisture_minmax'] = min_max_scaler.fit_transform(df_clean[['moisture_pct']])

#4.2. Z-Score Scaling (Standard Normalization)
std_scaler = StandardScaler()
df_clean['moisture_zscore'] = std_scaler.fit_transform(df_clean[['moisture_pct']])
df_clean[['moisture_pct', 'moisture_minmax', 'moisture_zscore']].head()
```

Out[4]:

	moisture_pct	moisture_minmax	moisture_zscore
0	6.229510	0.365153	-0.529272
1	6.525639	0.433927	0.003753
2	6.528640	0.434624	0.009155
3	6.439983	0.414034	-0.150426
4	7.116764	0.571212	1.067766

In [5]:
```
from sklearn.decomposition import PCA

#5. Dimensionality Reduction

# Select features for PCA
features = ['moisture_pct', 'temperature_C', 'humidity_pct']
x = df_clean[features]

# We must scale the data before PCA
x_scaled = StandardScaler().fit_transform(x)

# Apply PCA to reduce to 2 components
```

```python
pca = PCA(n_components=2)
principal_components = pca.fit_transform(x_scaled)

# Create a simple DataFrame for results
pca_df = pd.DataFrame(data=principal_components, columns=['PC1', 'PC2'])

print("Explained Variance Ratio:", pca.explained_variance_ratio_)
pca_df.head()
```

```
Explained Variance Ratio: [0.34200119 0.33581724]
```

Out[5]:

	PC1	PC2
0	0.625723	0.470008
1	-0.396366	0.799987
2	-0.847792	1.066519
3	1.716891	-3.602147
4	0.348700	-0.341943

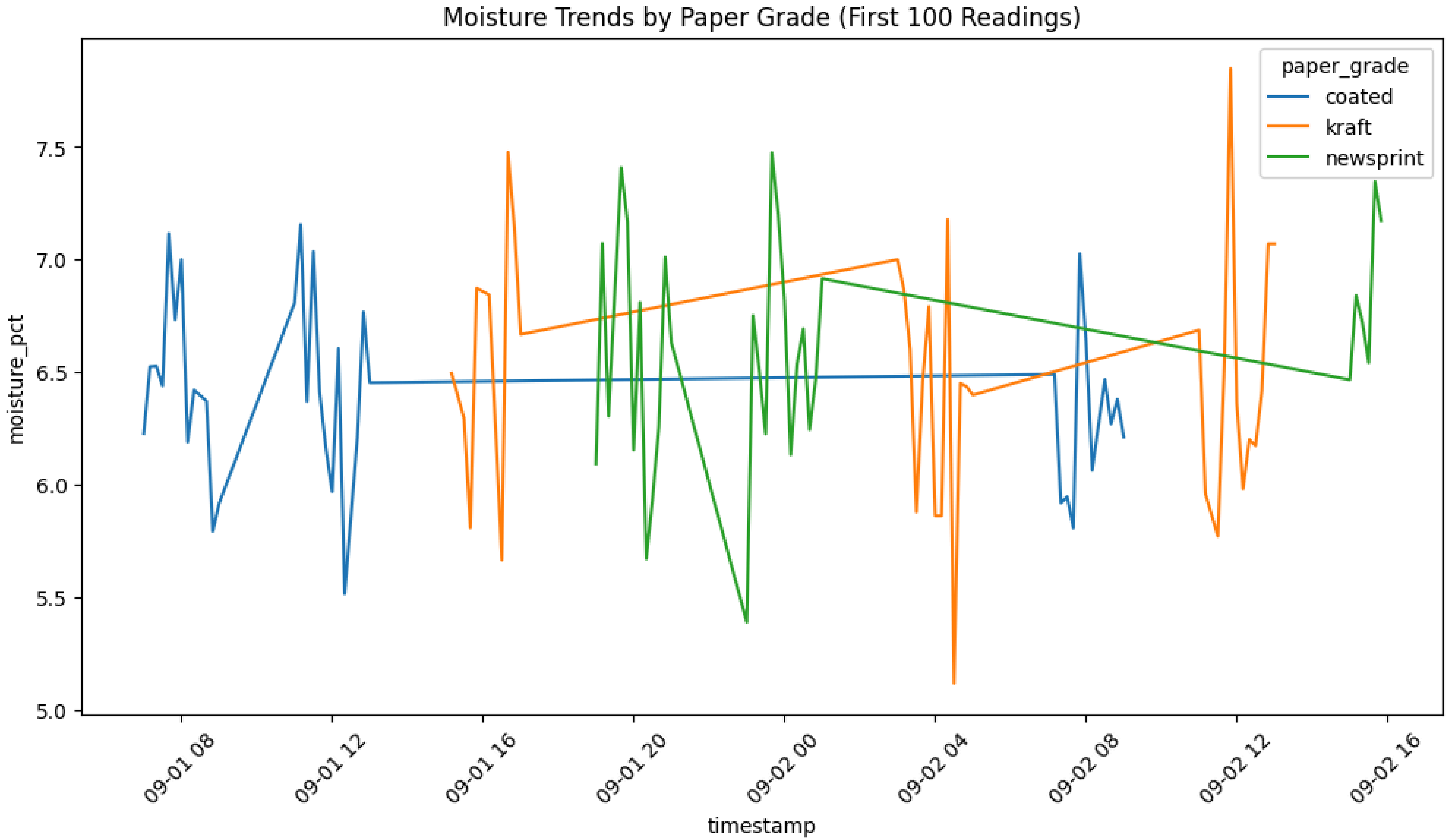
In [6]:

```python
from sklearn.decomposition import FactorAnalysis

#5.1 Apply Factor Analysis
fa = FactorAnalysis(n_components=2, rotation="varimax")
fa_components = fa.fit_transform(x_scaled)
print("Factor Analysis Run Successfully")
```

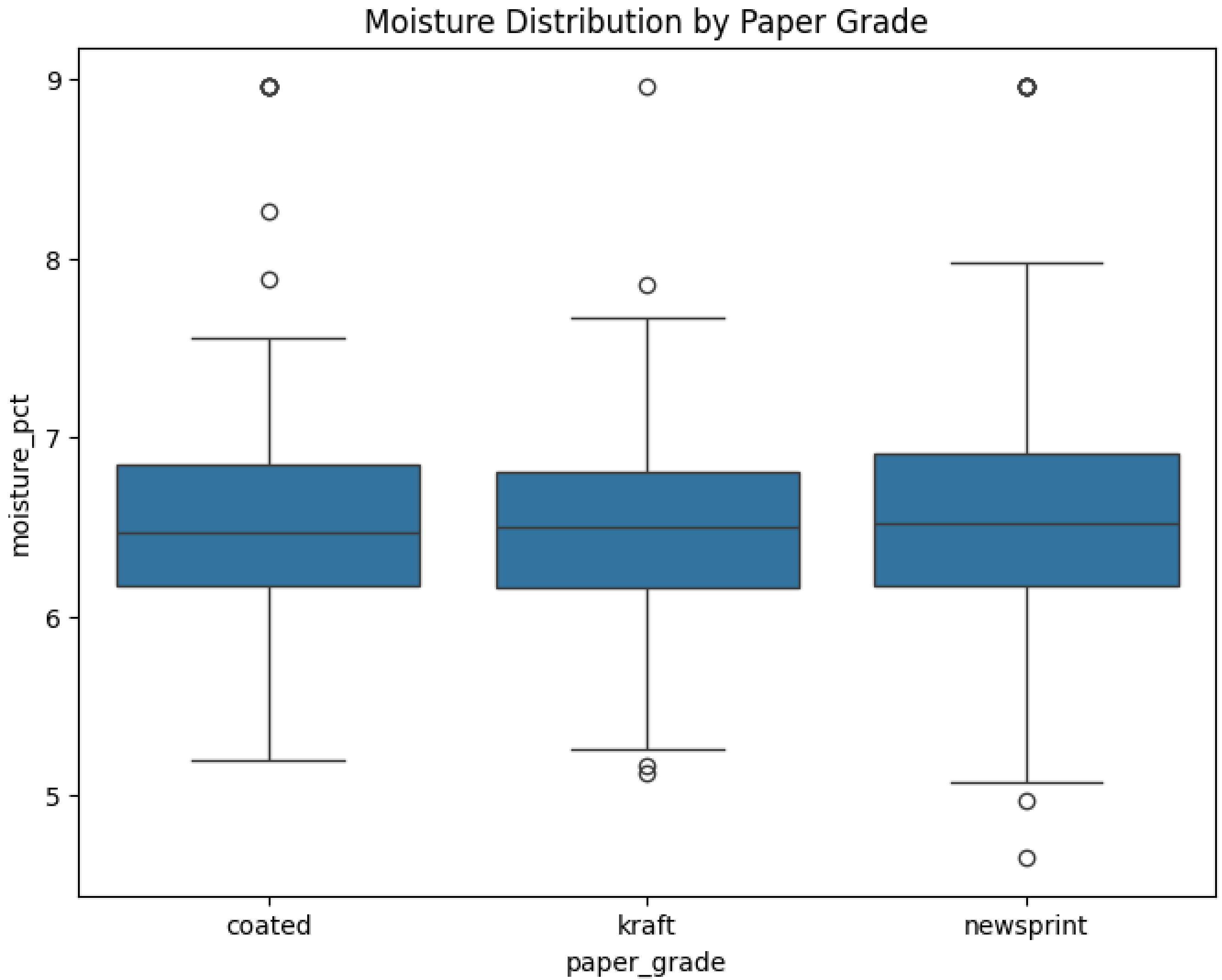
```
Factor Analysis Run Successfully
```

In [7]:

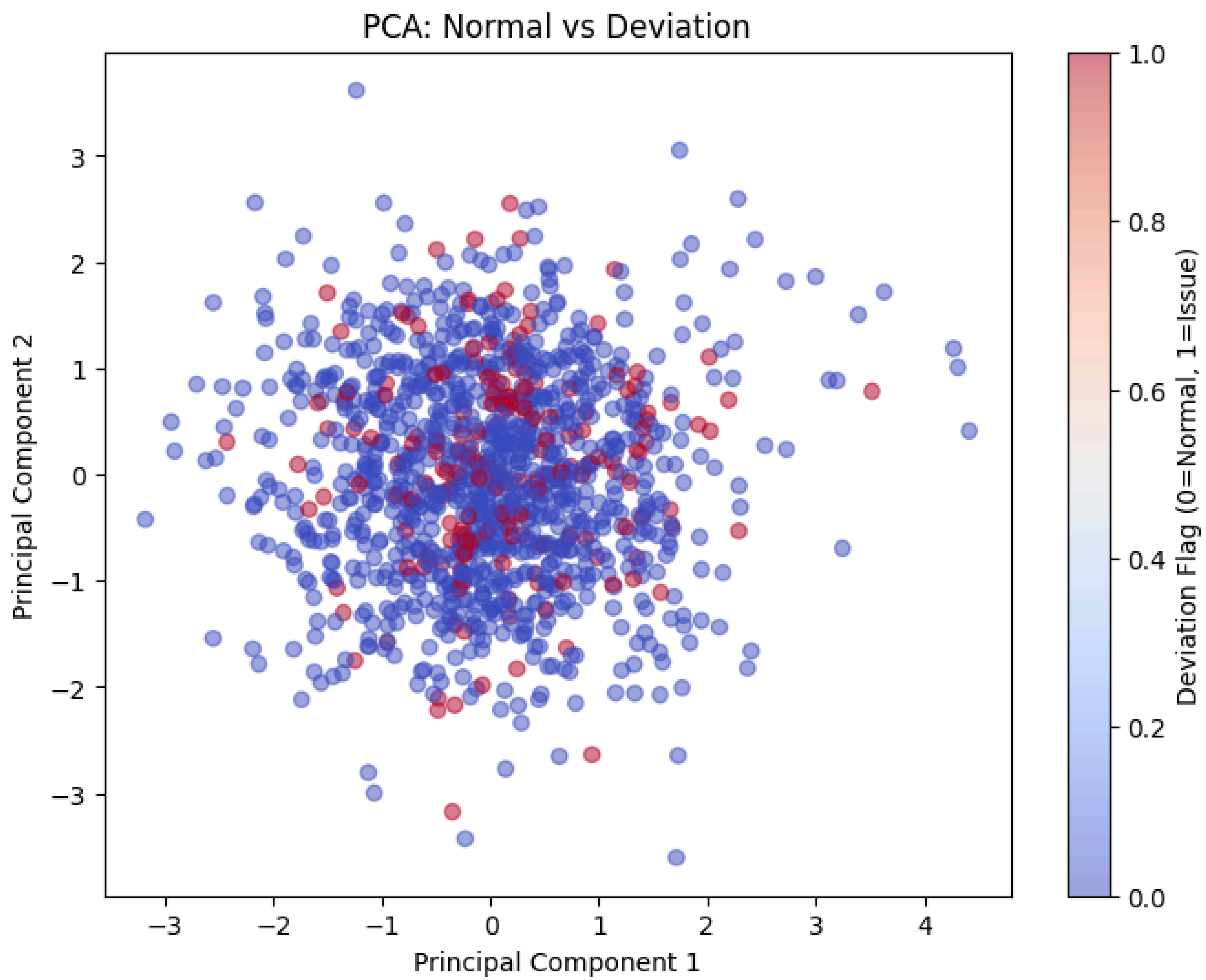
```python
import matplotlib.pyplot as plt
import seaborn as sns

#6. Visualization and Reporting

#6.1 Line Plot
plt.figure(figsize=(12, 6))
sns.lineplot(data=df_clean.iloc[:100], x='timestamp', y='moisture_pct', hue='paper_grade')
plt.title('Moisture Trends by Paper Grade (First 100 Readings)')
plt.xticks(rotation=45)
plt.show()
```



In [8]:
```
#6.2 Boxplot
plt.figure(figsize=(8, 6))
sns.boxplot(x='paper_grade', y='moisture_pct', data=df_clean)
plt.title('Moisture Distribution by Paper Grade')
plt.show()
```



In [9]:
```
#6.3 PCA results
plt.figure(figsize=(8, 6))
plt.scatter(pca_df['PC1'], pca_df['PC2'], c=df_clean['deviation_flag'], cmap='coolwarm', alpha=0.5)
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.title('PCA: Normal vs Deviation')
plt.colorbar(label='Deviation Flag (0=Normal, 1=Issue)')
plt.show()
```



In [10]:
```
#7. Data Integration and Export

df_clean.to_csv('final_paper_moisture_cleaned.csv', index=False)
print("CSV file saved successfully.")
```

```
CSV file saved successfully.
```

In [11]:
```
import sqlite3
```

```
#7.1 Data Integration
conn = sqlite3.connect('paper_project.db')
# Save dataframe to SQL table
df_clean.to_sql('moisture_data', conn, if_exists='replace', index=False)
# Close connection
conn.close()
print("Data exported to SQLite database.")
```

```
Data exported to SQLite database.
```

In [13]:

```
import sqlite3
import pandas as pd

#7.2 Connect to the database
conn = sqlite3.connect('paper_project.db')
# specific command to "Select All" from your table
query = "SELECT * FROM moisture_data"
#Read it into a pandas dataframe to view it
df_view = pd.read_sql(query, conn)
#Display
print(df_view.head())
conn.close()
```

```
             timestamp machine_id   batch_id paper_grade  moisture_pct  \
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   temperature_C  humidity_pct          cycle_start            cycle_end  \
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4      23.824238     45.115315  2025-09-01 07:00:00  2025-09-01 09:00:00

   deviation_flag  paper_grade_enc  machine_id_enc  moisture_minmax  \
0               0                0               3         0.365153
1               0                0               3         0.433927
2               0                0               3         0.434624
3               0                0               3         0.414034
4               0                0               3         0.571212

   moisture_zscore
0        -0.529272
1         0.003753
2         0.009155
3        -0.150426
4         1.067766
```

In [ ]: